Alpha-Phellandrene-Induced DNA Damage and Affect DNA Repair Protein Expression in WEHI-3 Murine Leukemia Cells *In Vitro*

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Received 10 March 2014; revised 28 April 2014; accepted 9 May 2014

ABSTRACT: Although there are few reports regarding α-phellandrene (α-PA), a natural compound from *Schinus molle* L. essential oil, there is no report to show that α-PA induced DNA damage and affected DNA repair associated protein expression. Herein, we investigated the effects of α-PA on DNA damage and repair associated protein expression in murine leukemia cells. Flow cytometric assay was used to measure the effects of α-PA on total cell viability and the results indicated that α-PA induced cell death. Comet assay and 4,6-diamidino-2-phenylindole dihydrochloride staining were used for measuring DNA damage and condensation, respectively, and the results indicated that α-PA induced DNA damage and condensation in a concentration-dependent manner. DNA gel electrophoresis was used to examine the DNA damage and the results showed that α-PA induced DNA damage in WEHI-3 cells. Western blotting assay was used to measure the changes of DNA damage and repair associated protein expression and the results indicated that α-PA increased p-p53, p-H2A.X, 14-3-3-σ, and MDC1 protein expression but inhibited the protein of p53, MGMT, DNA-PK, and BRCA-1. © 2014 Wiley Periodicals, Inc. Environ Toxicol 30: 1322–1330, 2015.

Keywords: α-phellandrene; WEHI-3 leukemia cells; DNA damage and repair

Supported by: China Medical University, Taichung, Taiwan (No. CMU100-ASIA-04). Experiments and data analysis were performed in part through the use of the Medical Research Core Facilities Center, Office of Research & Development at China medical University, Taichung, Taiwan, R.O.C.

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Published online 24 May 2014 in Wiley Online Library (wileyonlinelibrary.com). DOI: 10.1002/tox.22003

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INTRODUCTION

Although DNA damage is a relatively common event in the life of a cell, DNA damage can affect the genomic integrity of cells. It was reported that chromosomal double-strand break may arise from endogenous sources, including DNA replication and antigen receptor rearrangements in lymphocytes or arise from exogenous sources, for example the ionizing radiation (Lieber, 2010; Chapman et al., 2012; Kass et al., 2013) and in most cases, DNA damage came from exogenous sources (Zuo et al., 2012). It was well known that DNA is the major target for environmental genotoxins (Roos and Kaina, 2006) and eukaryotic cells have been shown to develop and involve many proteins in response to DNA damage recognition and repair (Zhou and Elledge, 2000).

Much evidence have shown that daily consumption of fruits and vegetables will decrease the risk of oxidative stress and its damage on cells (Bellion et al., 2010) and even can reduce the occurrence of human cancer (Katiyar et al., 1999; Hirao et al., 2010). *Schinus molle* L. have been reported to be used as antibacterial and topical antiseptic (Duke, 1985) and act as fungi toxicant (Dikshit et al., 1986). It was reported that the extract from the leaf of *Schinus molle* L. have antidepressant (Machado et al., 2007) and analgesic and central depressant (Barrachina et al., 1997) effects. The α-phellandrene (α-PA), the major components of *Schinus molle* L. essential oil (>50%) (Iscan et al., 2012), is one of the monoterpenes found in natural sources, and was often used in the food and perfume industry (Bendaoud et al., 2010), though the biological activity of α-PA is not available and not clear. Thus, in the present study, we investigated α-PA affecting the DNA damage and repairing associated proteins expression in leukemia cells *in vitro* and the results showed that α-PA induced DNA damage and affected DNA damage repair associated protein levels in mice leukemia WEHI-3 cells *in vitro*.

MATERIALS AND METHODS

Materials and Reagents

α-phellandrene (α-PA) and DMSO were purchased from Sigma-Aldrich (St. Louis, MO, USA). Fetal bovine serum (FBS), RPMI-1640 medium, L-glutamine and penicillin–streptomycin were purchased from Gibco Life Technologies (Carlsbad, CA, USA). α-PA was dissolved in DMSO at 0.1% in a tube covered with black paper to protect it from light and kept at −20°C until used in experiments.

Cell Culture and Treatment

The WEHI-3 murine leukemia cell line was obtained from the Food Industry Research and Development Institute (Hsinchu, Taiwan). The WEHI-3 cells were placed into 75 mm culture flask with RPMI 1640 medium containing 2 mM L-glutamine, supplemented with 10% heat-inactivated FBS and 100 U/mL penicillin and 100 μg/mL streptomycin and incubated in an incubator at 5% CO_2 and 37°C and 90% relative humidity. α-phellandrene was diluted in sterile PBS before adding to the culture (Chang et al., 2013).

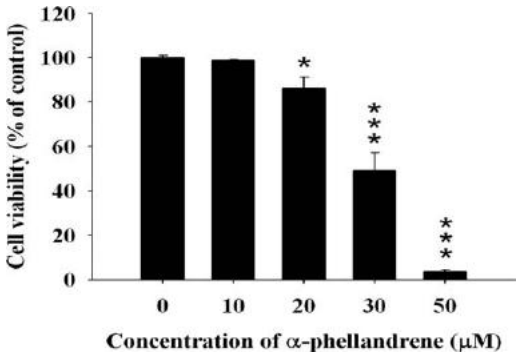

Fig. 1. α-PA affect the percentage of viable WEHI-3 cells. WEHI-3 cells (2×10^5 cells/well) were placed in 24-well plates and then were incubated with α-PA for 48 h. All cells were stained with PI (5 μg/mL) and analyzed by flow cytometry as described in "Materials and Methods". $^*P < 0.05$ and $^{***}P < 0.001$ were considered significant.

Cell Viability Analysis

WEHI-3 cells at the density of 2×10^5 cells/well in 24-well plates were maintained for 24 h in an incubator. Cells were incubated with α-PA (0, 10, 20, 30, and 50 μM) for 48 h. All cells in each well were collected and stained with propidium iodide (PI, Sigma-Aldrich) (5 μg/mL) and analyzed for total viable cells by flow cytometry (Becton-Dickinson, FACSCalibur, San Jose, CA, USA) and cell viability was calculated as previously described (Hsu et al., 2009).

Comet Assay for DNA Damage Measurement

WEHI-3 cells at the density of 2×10^5 cells/well in 24-well plates were incubated with α-PA (0, 10, 20, 30, and 50 μM) and 0.1% of hydrogen peroxide (H_2O_2, positive control) for 48 h in RPMI 1640 medium grown at 37°C in 5% CO_2 and 95% air. Cells were harvested to examine DNA damage using the Comet assay as described previously (Chueh et al., 2013; Hsu et al., 2012). The comet tail length were calculated using the TriTek CometScore™ software image analysis system (TriTek Corp., Sumerduck, VA, USA) as described previously (Hsu et al., 2012; Chueh et al., 2013).

4,6-Diamidino-2-phenylindole Dihydrochloride Staining for DNA Damage and Condensation

WEHI-3 cells (2×10^5 cells/well) were placed in six-well plates for 24 h and were treated with α-PA (0, 10, 20, 30,

and 50 μM) for 48 h. Then cells in the wells were stained by 4,6-diamidino-2-phenylindole dihydrochloride (DAPI) as described previously (Chiang et al., 2013; Chiu et al., 2013). At the end of staining, the cells were examined and photographed using a fluorescence microscope at 200×.

DNA Gel Electrophoresis for DNA Fragmentation Examinations

WEHI-3 cells (2×10^6 cells/well) in 10 cm dish were incubated with α-PA (0, 10, 20, 30, and 50 μM) for 48 h. Cells were collected and DNA was extracted from each treatment using DNA isolation kit (Genemark Technology, Tainan, Taiwan). The DNA electrophoresis was carried out at 15 volts in 1.5% agarose gel in Tris/Boric acid buffer for 2 h. At the end of electrophoresis, DNA was stained with ethidium bromide (EtBr, Sigma-Aldrich). The gel was examined and photographed under UV light box as described previously (Chiang et al., 2013; Chiu et al., 2013).

Western Blotting Assay for Protein Expression Examinations

WEHI-3 cells (2×10^6 cells) were placed in 10 cm dish and then were incubated with α-PA (0, 10, 20, 30, and 50 μM) for 48 h. Cells were harvested and lysed with lysis buffer containing 40 mM Tris–HCl (pH 7.4), 10 mM EDTA, 120 mM NaCl, 1 mM dithiothreitol, 0.1% Nonide P-40. All

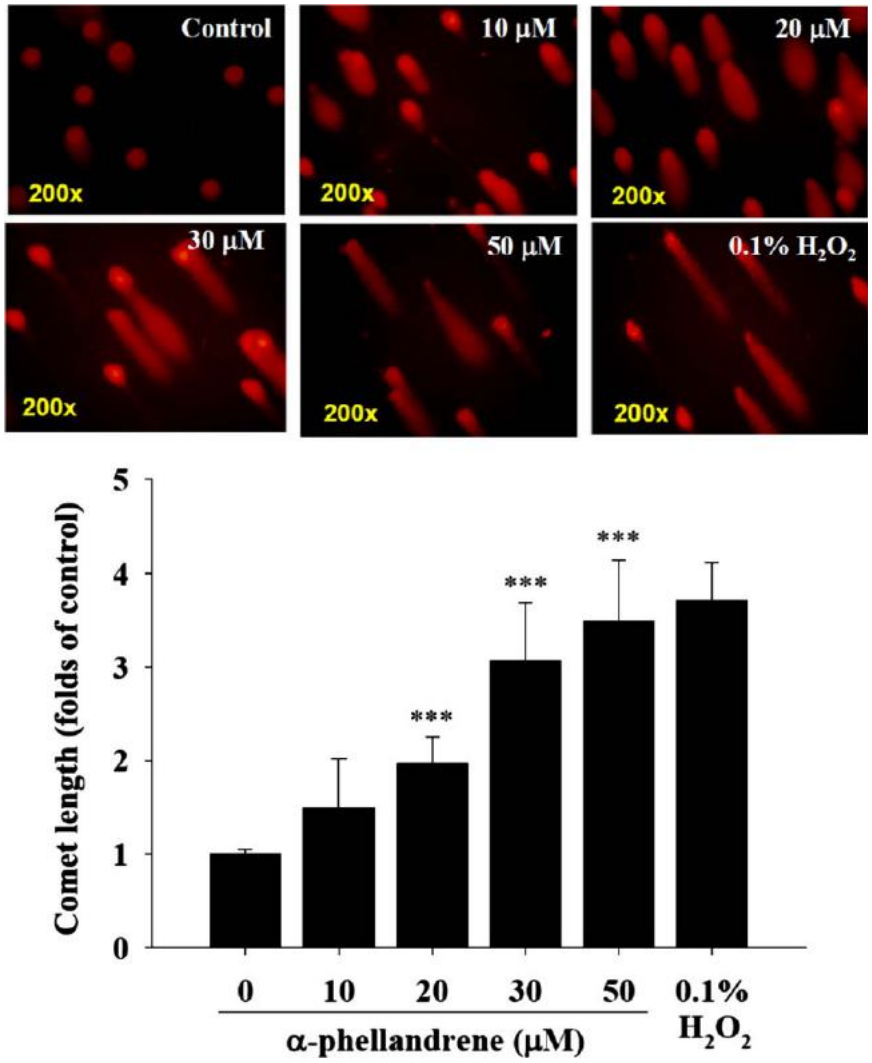

Fig. 2. α-PA induced DNA damage in WEHI-3 cells. WEHI-3 cells (2×10^5 cells/well) were placed in 24-well plates and then were incubated with 0, 10, 20, 30, and 50 μM of α-PA for 48 h and DNA damage was determined by Comet assay as described in "Materials and Methods". A: Representative picture of Comet assay for dose-dependent effects; B: Comet length (folds of control). The 0.1% H_2O_2 was acted as positive control. Arrow showing the comet tail (DNA damage). $^*P < 0.05$ and $^{***}P < 0.001$ were considered significant. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

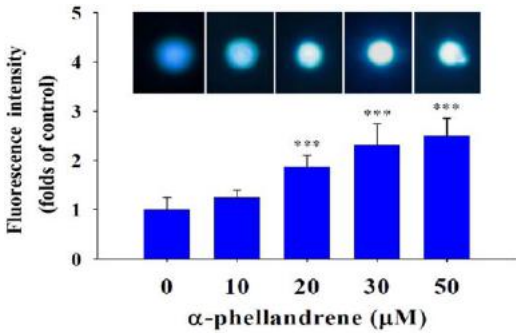

Fig. 3. α-PA promoted DNA condensation. WEHI-3 cells (2×10^5 cells/well) were placed in 6-well plates for 24 h and were treated with 0, 10, 20, 30, and 50 μM of α-PA for 48 h. Then cells were stained with 4,6-diamidino-2-phenylindole dihydrochloride (DAPI) as described in "Materials and Methods". At the end of staining, the cells were examined and photographed using a fluorescence microscope at 200×. $^*P < 0.05$ and $^{***}P < 0.001$ were considered significant. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

samples were measured for total protein by BioRad assay kit (Bio-Rad Laboratories, Hercules, CA, USA). 30 μg of protein from each treatment was loaded on a gel (10% Tris–glycine–SDS–polyacrylamide) for western blotting analysis. At the end of electrophoresis, all samples were transferred to a nitrocellulose membrane by electro-blotting. Primary antibodies for the DNA damage and repair associated proteins were individually used to stain each sample. Then all samples were washed and were followed by staining with secondary antibody for enhanced chemiluminescence (NEN Life Science Products, Boston, MA, USA) as described previously. Anti-β-actin (a mouse monoclonal antibody) was used as a loading control as described previously (Lu et al., 2009; Lai et al., 2013; Lin et al., 2013).

Confocal Laser Scanning Microscopy Assay for Protein Translocation

WEHI-3 cells (5×10^4 cells/ well) were placed on four-well chamber slides and incubated with 30 μM α-PA for 48 h and then were fixed in 4% formaldehyde in PBS for 15 min. This was followed by being permeabilized with 0.3% Triton-X 100 in PBS for 1 h then using 2% BSA for blocking of non-specific binding sites as described previously (Lu et al., 2009; Lai et al., 2013). Each sample was stained with primary antibodies to p-H2A.X, p-p53 and MDC1 (1:100 dilution) (green fluorescence) overnight before being washed twice with PBS then being stained with secondary antibody (FITC-conjugated goat anti-mouse IgG) (1:100 dilution) and were stained by PI (red fluorescence). All samples were photomicrographed by using a Leica TCS SP2 Confocal Spectral Microscopy (Lu et al., 2009; Lai et al., 2013).

Statistical Analysis

All data were expressed as the mean ± SD from three experiments. Student's *t* test was used to analyze differences between positive control and α-PA-treated groups. All the statistical analyses were performed $^*P < 0.05$, which was considered significant.

RESULTS

α-PA Effects on the Percentage of Viable WEHI-3 Cells

Cells were treated with α-PA for 48 h and then were harvested for determining the total viability by using flow cytometry assay and the results are shown in Figure 1. Results indicated that α-PA significantly decreased the total viable cells when compared with the control group.

α-PA-Induced DNA Damage in WEHI-3 Cells

Cells were treated with various doses of α-PA for 48 h and then were measured for DNA damage by using Comet assay and the results are shown in Figures 2(A) and (B). Results indicated that α-PA induced DNA damage based on the

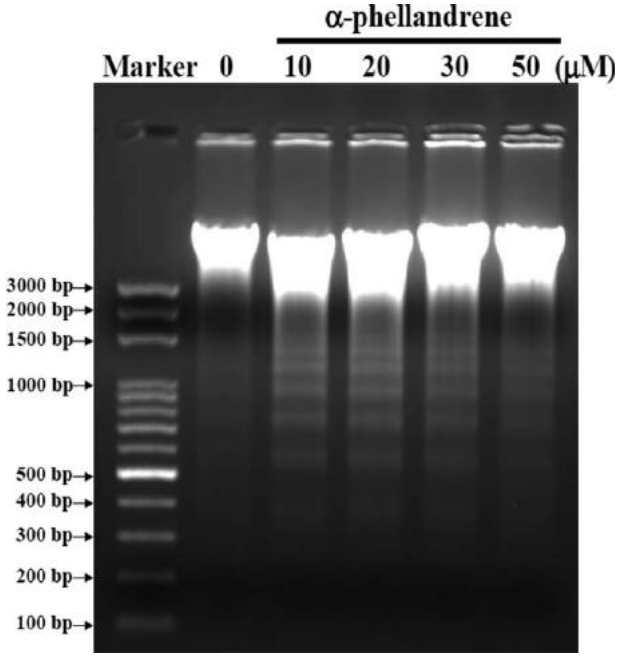

Fig. 4. α-PA-induced DNA damage was examined by DNA gel electrophoresis. WEHI-3 cells (2×10^5 cells/well) were placed in 10 cm dish and were treated with 0, 10, 20, 30, and 50 μM of α-PA for 48 h. Then cells were collected and DNA was isolated from each treatment and then we performed DNA gel electrophoresis as described in "Materials and Methods".

production of Comet tail and the higher doses of α-PA led to the longer Comet tail. Based on these observations, it indicated that α-PA induced DNA damage in a dose-dependent manner.

α-PA-Induced DNA Condension in WEHI-3 Cells

Cells were treated with various doses of α-PA for 48 h and then were measured for the DNA condensation using DAPI stain assay and the results are shown in Figures 3(A) and (B). Results indicated that the higher doses of α-PA induced DNA condensation based on the production of higher staining of DAPI. Based on these observations, it indicated that α-PA induced DNA condensation in a dose-dependent manner.

α-PA-Induced DNA Fragmentation in WEHI-3 Cells

Cells were treated with various doses of α-PA for 48 h and then were measured for the DNA fragmentation by using DNA gel electrophoresis assay and the results are shown in Figure 4. Results indicated that α-PA induced DNA fragmentation in a dose-dependent manner.

α-PA Effects on DNA Damage and Repair Associated Protein in WEHI-3 Cells

Cells were treated with 0, 10, 20, 30, and 50 μM of α-PA for 48 h and then were harvested and measured for the protein expression using western blotting assay and the results are shown in Figures 5(A) and (B). Results indicated that α-PA decreased the expression of p53, MGMT, DNA-PK, and BRCA-1 but increased the expression of p-p53, p-H2A.X, 14-3-3σ, and MDC1 that indicated that α-PA induced DNA damage may be through the inhibition of repair protein expression.

α-PA Affects p-p53, p-H2A.X, and MDC1 Protein Translocation in WEHI-3 Cells

Cells were treated with 30 μM of α-PA for 48 h and then were examined for the translocation of p-p53, p-H2A.X, and MDC1 protein using confocal laser microscopic assay and the results are shown in Figures 6(A–C). Results indicated that α-PA inhibited the translocation of p-H2A.X [Fig. 6(A)] and MDC1 [Fig. 6(C)] from cytosol to nuclei but promoted the translocation of p-p53 [Fig. 6(B)] from cytosol to nuclei (Fig. 6).

DISCUSSION

Reports of α-PA on biological activity are rare and reports also show that α-PA did not induce significant antimicrobial activity. However, there is no report to show that α-PA induced DNA damage in cancer cells. In the present study, we investigated the effects of α-PA on DNA damage and repair associated protein expression in murine leukemia cells *in vitro*. Herein, we found that α-PA induced significant cell death at 10, 20, 30, and 50 μM when compared to control groups. Based on examinations of Comet assay and DAPI staining, it indicated that α-PA induced DNA damage and condensation, respectively. We also used DNA gel electrophoresis which showed that α-PA induced DNA fragmentation in WEHI-3 cells. Furthermore, we used western blotting which indicated that α-PA inhibited DNA damage and repair associated protein expression and confocal laser microscopic assay also confirmed that α-PA affect p-p53, p-H2A.X, and MDC1 protein translocation in WEHI-3 cells.

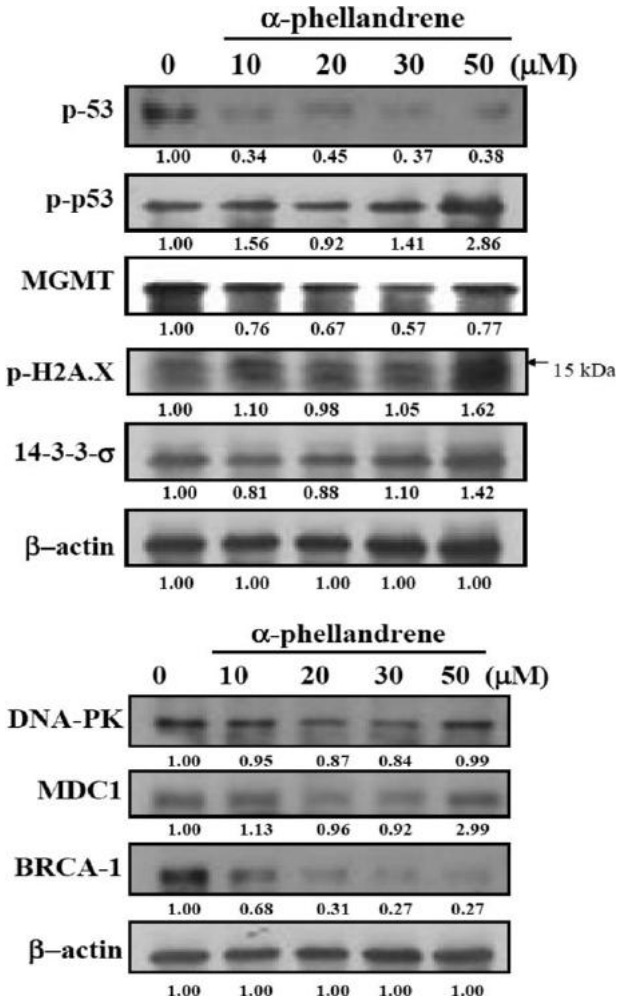

Fig. 5. Western blotting assay for protein levels associated with DNA damage and repair in WEHI-3 cells. Cells (2×10^6 cells/dish) were placed in 10 cm dish then were incubated with 0, 10, 20, 30, and 50 μM of α-PA for 48h. The total proteins were determined by Bio-Rad assay Kit. The amount of proteins were measured by sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) and immunoblotting as in "Materials and Methods".

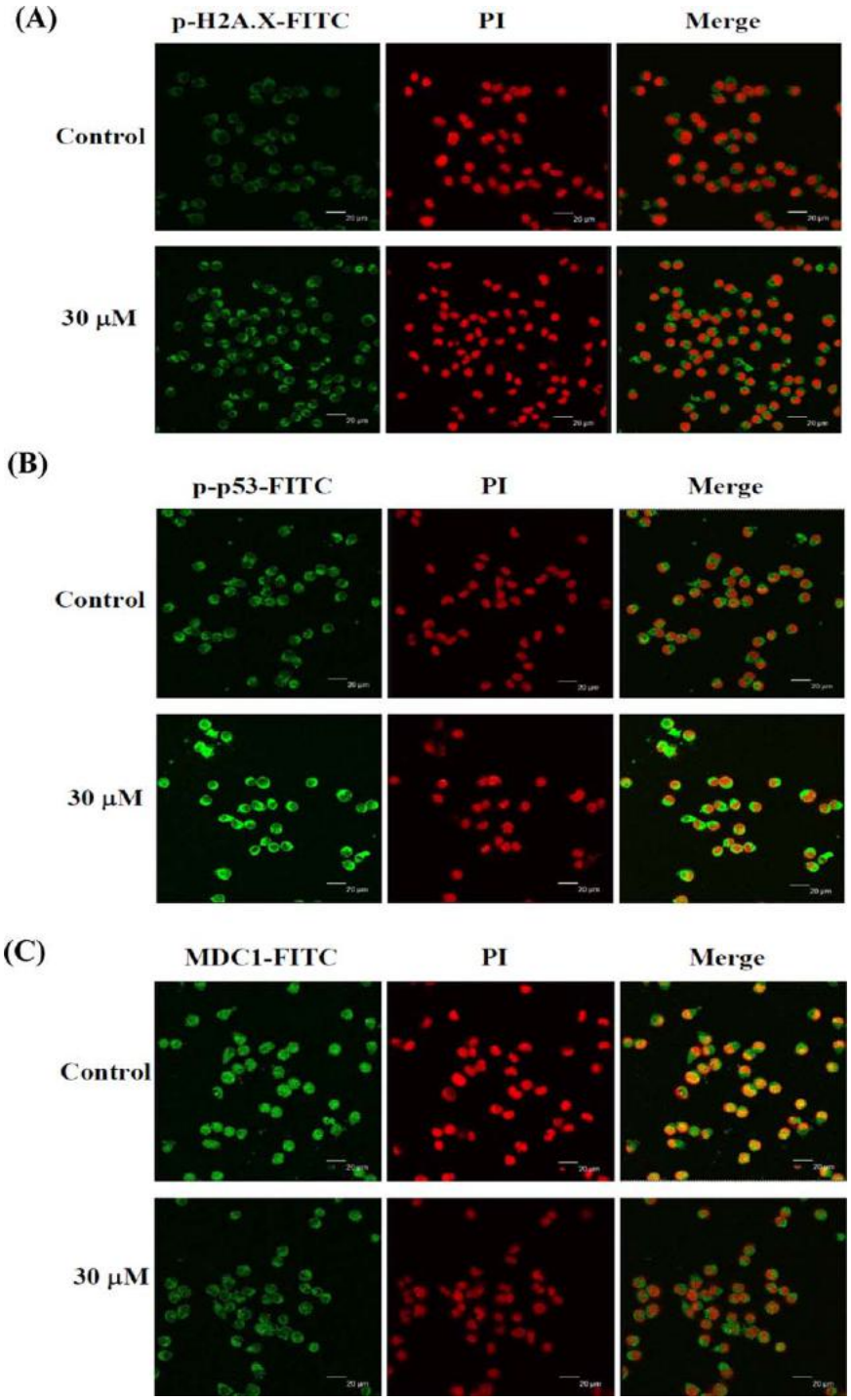

Fig. 6. Confocal laser system microscopy was used for examining protein translocation in WEHI-3 cells. Cells (5×10^4 cells/well) were placed on four-well chamber slides and incubated with 0 and 30 μM of α-PA for 48 h and then were fixed in 4% formaldehyde in PBS for 15 min followed by using 0.3% Triton-X 100 in PBS for 1 h then followed by immunostaining as described in Materials and Methods. A: p-H2A.X; B: p-p53 C: MDC1 examined and photomicrography under a Leica TCS SP2 Confocal Spectral Microscope. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

It is well known that the endogenous and exogenous genotoxic agents can induce damage DNA, compromise its normal function and lead to instable genome function or cause cell death. After cells induce DNA damage by agents, DNA repair to the damaged gene is critical for cells to live, thus numerous reports have shown the role of DNA repair system

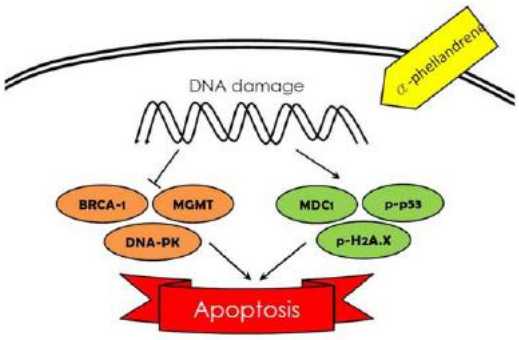

Fig. 7. Possible flow chart for α-PA-induced DNA damage and affecting DNA repair associated proteins in murine leukemia WEHI-3 cells. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

on DNA lesions (Olive et al., 1990; Tice et al., 1990). In order to further investigate the effects on DNA damage and repair associated protein expression, we used Western blotting to examine the levels of ATM, ATR, BRCA-1, 14-3-3σ, DNA-PK, MGMT, p-p53, p-H2A.X, and MDC1 in examined WEHI-30 cells. Results showed that α-PA decreased the protein levels of p53, MGMT, DNA-PK, and BRCA-1 but increased the levels of p-p53, p-H2A.X, 14-3-3 σ, and MDC1.

Results from Figure 5 have shown that α-PA increased the expression of p-p53; p53 increase is a hall marker of DNA damage in cells (Heo et al., 2012; Wu et al., 2013), however, we did not see the increase of p-53 but we did see the increase of p-p53. It was reported that *ataxia telangiectasia* mutated (ATM) and ATM- and Rad3-related (ATR) kinases are two master regulators of DNA damage signaling in human cells (Ciccia and Elledge, 2010). Results also showed that α-PA promoted the expressions of ATM and ATR proteins. This is in agreement with other reports which showed that agents induced double-stranded DNA breaks which can activate two master checkpoint kinases (ATM and ATR) (Choi et al., 2009; Shiotani and Zou, 2009) for cells keeping genomic integrity. Figure 6 showed that α-PA inhibited the tumor suppressor BRCA-1 protein expression in WEHI-3 cells. The BRCA1 protein is multifunctional including the maintenance of genomic stability (Huen et al., 2010); in breast and ovarian cancer, BRCA1 plays an important role in DNA repair, cell cycle checkpoint control and maintenance of genomic stability (Venkitaraman, 2002).

We also found that α-PA promoted the expressions of p-H2A.X but inhibited the expression of 14-3-3σ and DNA-dependent protein kinase (DNA-PK) in WEHI-3 cells. H2AX was involved in the efficient accumulation of DNA repair factors at the break site, and H2AX-deficient mice have been reported to have higher radio sensitivity (Celeste et al., 2002, 2003). The 14-3-3 σ was suggested to act as an effective therapeutic target in breast cancer based on the 14-3-3 σ overexpression as a novel molecular marker for disease recurrence (Neal et al., 2009). DNA-PK protein is involved and plays an important role in DNA damage repair (Dejmek et al., 2009; Mi et al., 2009).

We also found that α-PA inhibited the levels of DNA-PK and O^6-methylguanine DNA methyltransferase (MGMT) [Fig. 5(B)]. DNA-PK was reported in cells after a sequential activation of DNA-PK and PARP-1 will arrest S phase of cell cycle by DNA damage (Allay et al., 1999) and sub-nuclear induction of complex DNA damage may elicit the pan-nuclear activation of ATM and DNA–PK (Calkins et al., 2013). Numerous studies from mice have shown that tumorigenic and mice overexpressed human MGMT can selectively reduce O^6methylguanine mediated carcinogenic mutations to threshold levels after *N*-methyl-*N*-nitrosourea (Meyer et al., 2013) and it also can act against methylating agent induced tumor formation (Dumenco et al., 1993; Zaidi et al., 1995). Taken together, α-PA induces cell death may be through the induction of DNA damage via the inhibition of DNA repair associated protein expressions including such as MGMT, DNA-PK, and BRCA-1 for causing DNA damage in WEHI-3 cells *in vitro* (Fig. 7).

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